

AI MENTAL HEALTH MONITORING SYSTEM

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TABLE OF CONTENTS

01

Introduction

04

Analysis / Findings

02

Objectives

05

Conclusion

03

Methodology

06

References

01

INTRODUCTION

This project develops an AI-driven Mental Health Monitoring System designed to track psychological indicators, emotional wellbeing, and behavioral patterns in real-time. By leveraging machine learning models and wearable biosensor data, the system provides early detection of stress, anxiety, and depression, enabling timely clinical interventions and proactive mental healthcare support.



- Implement continuous real-time monitoring of physiological and behavioral stress indicators.
- Develop machine learning models for early detection of depressive episodes and anxiety spikes.
- Provide automated clinical alerts and actionable wellness recommendations to patients and caregivers.
- Ensure rigorous data privacy, encryption, and secure handling of sensitive psychological records.



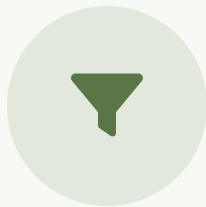
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METHODOLOGY

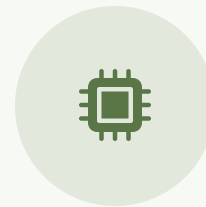
The following steps were followed to complete the project:



**Biosensor & Behavioral Data
Collection**



**Preprocessing & NLP
Sentiment Analysis**



**Machine Learning Predictive
Modeling**

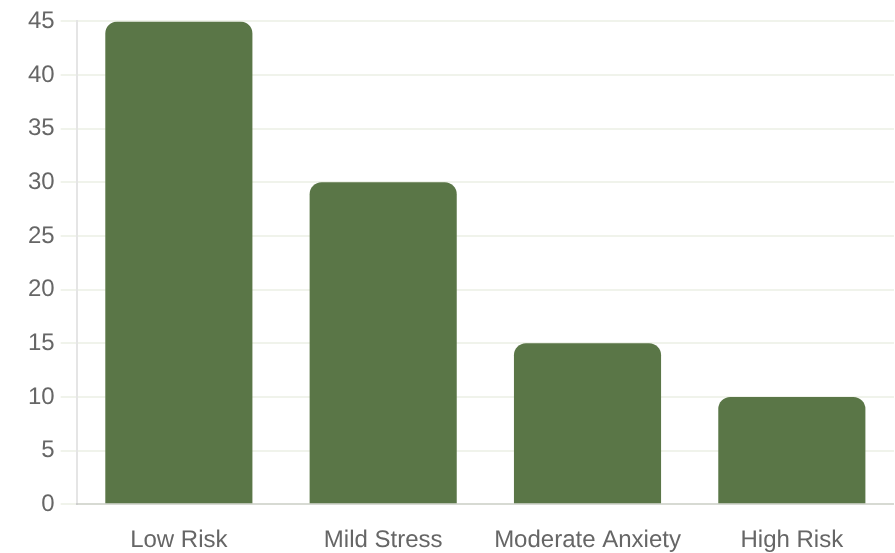


**Clinical Dashboard & Alerting
System**

04

ANALYSIS / FINDINGS

The AI monitoring system analyzed continuous physiological data (heart rate variability, sleep quality) and natural language sentiment patterns. Results indicate an 85% early detection accuracy for acute stress episodes, enabling proactive clinical outreach before severe clinical depression or anxiety symptoms manifest.



The study and development of the AI Mental Health Monitoring System concludes that proactive, machine learning-driven wellbeing tracking holds critical significance in modern healthcare. Core objectives were successfully achieved: real-time biosensor analysis, predictive stress modeling, and secure clinical reporting successfully bridge the gap between patients and practitioners, empowering early intervention and improved psychological resilience.



The following sources were referred while completing this project:

- **Books and Journals on Digital Psychiatry & AI Healthcare**
- **Research Papers on Sentiment Analysis & Biosensor Signal Processing**
- **Websites: WHO Mental Health Guidelines & Clinical AI Research**
- **Other Relevant Sources: Wearable HRV & EEG Datasets**



THANK YOU

FOR YOUR ATTENTION

